

Phone Detection using CNN Models

Under the supervision of:

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Problem Statement

Phones have become a tool which people use every minute of their waking life which unfortunately also includes the time they're behind the steering wheels of a car. This has increased the number of distracted drivers on the road, be it texting or being on a phone call.

Our goal is to produce a model which is reliable enough to be used in detecting the drivers who are distracted by their phone.

Methodology

The Statefarm distracted drivers dataset was used, only selecting the images classified as "normal driving" as safe, "texting - right" and "texting - left" as texting and "talking on the phone - right" and "talking on the phone - left" as phone. Afterwards data augmentation was done to prevent overfitting.

Two prebuilt models were used and tuned, those being MobileNetV2 and EfficientNet.

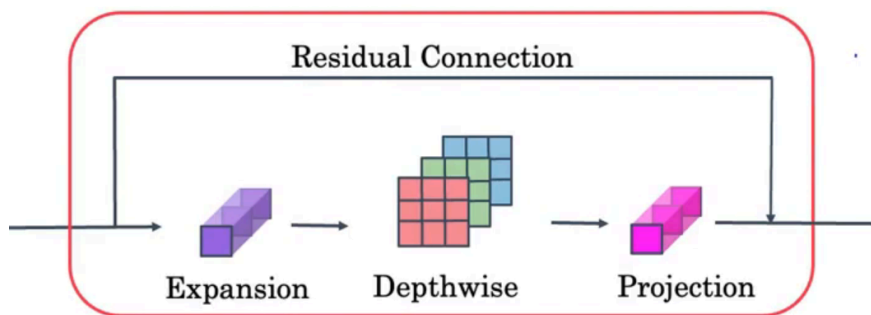
MobileNetV2

Theory

Nicknamed "ML on your phone", MobileNet is a family of CNN architectures which's main goal is to be lightweight.

This is achieved by using depth separable convolution which is split into 2 steps which is Depthwise convolution, instead of applying a 3D filter across all input channels simultaneously, a single 2D filter is applied to each input channel independently. For an RGB input image, the depthwise layer generates exactly 3 and Pointwise Convolution To combine and synthesize the feature maps isolated by the depthwise step, a 1x1 standard. convolution is applied next. This step projects and merges the isolated channel data into a specified channel space (output dimension), driving down computation while preserving cross-channel correlation.

To improve on the original model, V2 adds to structure to make it more accurate those introduces inverted residual blocks which are made up of expansion, depthwise and projection layers with a residual skip



Implementation

Driver-Based Splitting:

- prevent the same driver from appearing in both training and validation.

Image Preprocessing:

- MobileNet preprocessing was applied using: `preprocess_input` to improve feature extraction and improve webcam consistency.

Data Augmentation:

- Training images were augmented using: rotation, zoom, shifting and brightness adjustment to reduce overfitting .

Creating Image Generators:

- training generator and validation generator.

Building MobileNetV2 Model:

- To use pretrained visual features learned from ImageNet.

Building the Classification Head:

- Using GlobalAveragePooling2D, Dropout(0.5) and Dense(3, softmax) to adapt MobileNetB2 for driver behavior classification.

Compiling the Model:

- The model was compiled using: Adam optimizer, categorical crossentropy loss, accuracy metric.

Setting Training Callbacks:

- Three callbacks were used: EarlyStopping and ModelCheckpoint.

Initial Training Phase:

- The base MobileNet layers were frozen, to preserve pretrained ImageNet features while training the classifier.

Fine-Tuning Phase:

- The last 30 MobileNet layers were unfrozen, to adapt high-level features to phone usage, hand positions and driver posture

Model Evaluation:

- The trained model was evaluated using: validation accuracy, validation loss

EfficientNetB2

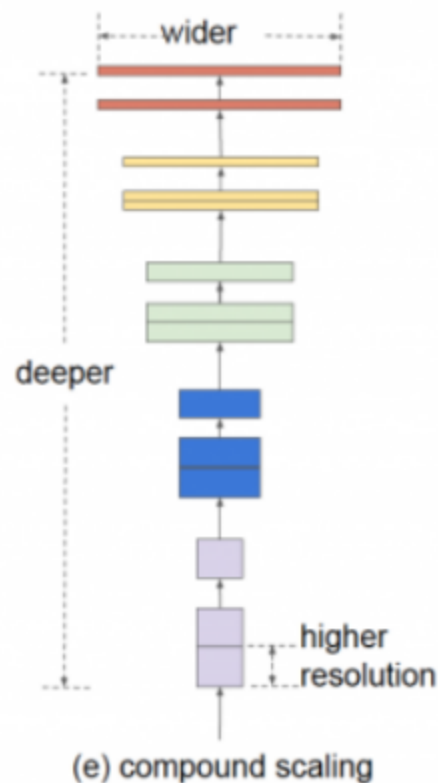
Theory

Nicknamed an “accuracy-efficient CNN,” EfficientNet is a family of convolutional neural networks designed to achieve strong accuracy while keeping computation relatively low. This is mainly achieved through compound scaling, where the model’s depth, width, and input resolution are scaled together in a balanced way rather than increasing only one dimension.

EfficientNetB2 is a scaled-up version of the baseline EfficientNetB0. It is larger and more accurate than B0 and B1, but still much more efficient than many older CNNs. Its core building block is the MBConv block, which is inspired by MobileNetV2’s inverted residual structure.

The MBConv block begins with an expansion layer, where a 1×1 convolution increases the number of channels to give the network more feature space. This is followed by a depthwise convolution, where each channel is filtered independently to reduce computation. EfficientNet then adds a Squeeze-and-Excitation module, which helps the model emphasize the most important channels and suppress less useful ones. Finally, a projection layer uses a 1×1 convolution to reduce the channels back down to a compact representation. When possible, a residual skip connection is added to improve information flow and make training more stable.

Overall, EfficientNetB2 improves on simpler CNNs by combining MobileNet-style efficient convolutions, attention through squeeze-and-excitation, and balanced compound scaling, allowing it to achieve strong image classification performance without becoming too computationally expensive.



Steps of implementation:

Driver-Based Splitting:

- prevent the same driver from appearing in both training and validation.

Image Preprocessing:

- EfficientNet preprocessing was applied using: preprocess_input to improve feature extraction and improve webcam consistency.

Data Augmentation:

- Training images were augmented using: rotation, zoom, shifting and brightness adjustment to reduce overfitting .

Creating Image Generators:

- training generator and validation generator.

Calculating Class Weights:

- To balance learning between classes.

Building EfficientNetB2 Model:

- To use pretrained visual features learned from ImageNet.

Building the Classification Head:

- Using GlobalAveragePooling2D, Dropout(0.5) and Dense(3, softmax) to adapt EfficientNetB2 for driver behavior classification.

Compiling the Model:

- The model was compiled using: Adam optimizer, categorical crossentropy loss, accuracy metric.

Setting Training Callbacks:

- Three callbacks were used: EarlyStopping, ReduceLROnPlateau and ModelCheckpoint.

Initial Training Phase:

- The base EfficientNet layers were frozen, to preserve pretrained ImageNet features while training the classifier.

Fine-Tuning Phase:

- The last 30 EfficientNet layers were unfrozen, to adapt high-level features to phone usage, hand positions and driver posture

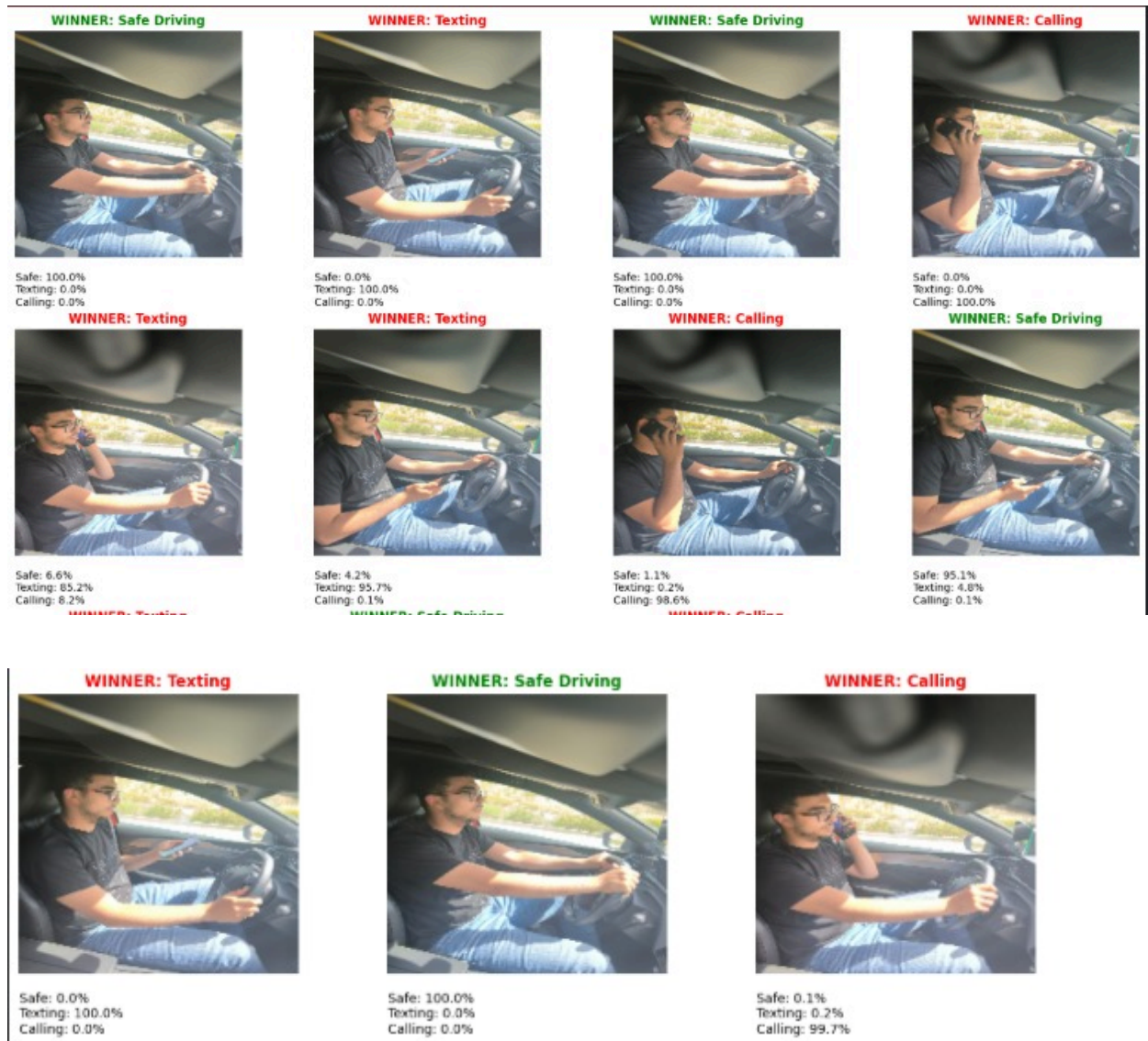
Model Evaluation:

- The trained model was evaluated using: validation accuracy, validation loss and confusion matrix

Conclusion

Results

MobileNetV2



AI Biases

- In the original dataset drivers of many different races were included to ensure fairness but it had an issue where drivers weren't tasked equally. But races weren't included equally.
- Model was trained over the driver seats angle. Model will fail if used over shiny windows. Where person might not be visible.

- Our Models are more sustainable due to using light weight models which would consume less energy
- Misuse and invasion of privacy as models take pictures of drivers.

Contribution Log

Moaaz Saleh	23-101263	Deployment of MobileNet
Mohamed Tarek	23-101248	Tuning of MobileNet
Hana Medhat	23-101222	Tuning of EfficientNet
Farah Sultan	23-101224	Deployment of EfficientNet